

Luke Chamandy

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Primary Interests

- Interacting binary stars, with emphasis on common envelope evolution.
 - Methods include global 3D hydrodynamical simulations with adaptive mesh refinement.
 - Key works include: LC et al. 2018; LC et al. 2019b; LC et al. 2020; LC et al. 2024; Bhattacharyya, LC, et al. 2026.
- Galactic magnetic fields and MHD, with emphasis on dynamos and comparison with observational data.
 - Numerical, semi-analytic and analytic methods to solve non-linear mean-field dynamo equations.
 - Key works include: LC 2016; Rodrigues, LC et al. 2019; Beck, LC et al. 2019; LC & Shukurov 2020; Jose, LC et al. 2024; Ghosh, LC et al. 2026.
- Exoplanets, with emphasis on the origin of white dwarf planets.
 - Mainly analytic estimates to explore this nascent field: LC et al. 2021; LC et al. 2024.

Positions Held

Faculty Positions

Jun 2022–Present: Reader, National Institute of Science Education and Research, Bhubaneswar, India

Postdoctoral Positions

Oct 2016–Jun 2022: Postdoctoral Associate, University of Rochester, USA

Oct 2014–Sep 2016: Postdoctoral Fellow, University of Cape Town, South Africa

Courses Taught/Teaching Positions

Jan–May 2026: PHY102: Introd. physics II (electrodynamics, special relativity, modern physics), NISER
Aug–Dec 2025: PHY101: Introductory physics I (mechanics and thermal physics, with calculus), NISER
Jan–May 2025: PHY141: Introductory physics laboratory, NISER
Aug–Dec 2024: P463: Astronomy and Astrophysics (upper-year B.Sc/M.Sc-level course), NISER
Jan–May 2024: P464: Plasma Physics and MHD (upper-year B.Sc/M.Sc-level course), NISER
Aug–Dec 2023: P345: Optics laboratory (mid-year B.Sc-level course, NISER)
Aug–Dec 2022: P463: Astronomy and Astrophysics (upper-year B.Sc./M.Sc.-level course), NISER
Oct–Dec 2013: Teaching Assistant, Inter-University Centre for Astronomy and Astrophysics, Pune
Aug 2006–Jul 2009: Physics Teacher (IB diploma), Mahindra United World College of India, Pune
Aug 2004–May 2005: Science and Mathematics Teacher, Jaanimmarik School, Kuujjuaq, Canada
Aug 2001–May 2002: Teaching Assistant, Dept. of Astronomy and Astrophysics, University of Toronto
Jan 2000–May 2001: Teaching Assistant, Dept. of Physics, Queen’s University, Canada

Education

- Jul 2009–Sep 2014: Ph.D., Inter-University Centre for Astronomy and Astrophysics, Pune, India
Doctoral dissertation: *The Origin of Large-scale Magnetic Fields in Galaxies*
Thesis Advisor: Prof. Kandaswamy Subramanian
Degree awarded: April 13, 2015 (awarding institute: Jawaharlal Nehru University)
- Aug 2005–May 2006: B.Ed., Ontario Institute for Studies in Education, University of Toronto
Teaching subjects: Physics, General Science
Specialization: Gifted Education
- Aug 2001–Mar 2003: M.Sc., Astrophysics, University of Toronto
Project Advisor: Prof. Christopher Thompson
- Jun 2000–May 2001: B.A. (Minor), Philosophy, Queen’s University, Canada
- Aug 1997–May 2000: B.Sc. (Honours), Physics, Queen’s University, Canada
Subject of Specialization: Astrophysics
Honours Thesis Advisor: Prof. Kayll Lake
- Aug 1995–May 1997: D.E.C., Physical Sciences, Marianopolis College, Montreal

Supervision of Students

- Jan 2023–Present: Mr. Sukanta Ghosh, Ph.D., NISER
Modeling the Radio-continuum of Galaxy Populations Evolving Over Cosmic Time
- Aug 2024–Present: Mr. Shubhajit Saha, Ph.D, NISER
Bridging 3D Simulations and 1D Simulations of Common Envelope Evolution
- Aug 2025–Present: Ms. Baishali Sur, M.Sc., NISER; has now begun a PhD at NISER
Direct Comparison of 3D and 1D Simulations of Common Envelope Evolution
- Aug 2026–Present: Ms. Sri Uma Maheshwari, M.Sc., NISER
Effect of Outflows on the Evolution of Galactic Magnetic Fields
- Aug 2026–Present: Mr. Shiva Thanay Pandiri, M.Sc., NISER
Observational Signatures of Common Envelope Events
- Aug 2025–May 2026: Mr. Shishir Halaharivi, M.Sc., NISER
Large-scale Magnetic Field Reversals and their Evolution in Galaxies
Current position: Ph.D. at Laboratoire d’Astrophysique de Marseille, France
- Aug 2024–May 2025: Ms. Gayathri Kalangadan, M.Sc., NISER
The Role of the Vishniac Magnetic Helicity Flux in Galactic Dynamos
Current position: Ph.D. at Leibniz Institute for Astrophysics Potsdam, Germany
- Aug 2024–May 2025: Ms. Upasana Das, M.Sc., NISER
Hydrodynamical Simulations of an Interacting Circumbinary Companion with the Inner Binary Undergoing Common Envelope Evolution
Current position: Ph.D. at Institute of Astronomy, KU Leuven, Belgium
- Aug 2023–May 2024: Ms. Abha Vishwakarma, M.Sc., NISER
Gravitational Waves from Neutron Stars Undergoing Common Envelope Evolution
Current position: Ph.D. at University of Illinois, Urbana-Champaign, USA
- Aug 2023–May 2024: Ms. Gayathri Santhosh, M.Sc., NISER
Explaining the Magnetic Fields of Nearby Galaxies Using Turbulent Dynamo Models
Current position: Ph.D. at INAF-IRA, University of Bologna, Italy
- Aug 2023–May 2024: Mr. R. Vasanth Kashyap, M.Sc., NISER (jointly supervised with Tuhin Ghosh)
Multiwavelength Observations of Galaxies and Their Magnetic Fields
Current position: Ph.D. at Scuola Normale Superiore di Pisa, Italy
- Jan 2023–May 2024: Mr. Deepanshu Gurjar, M.Sc., NISER
The Impact of Neutron Star Jets in Common Envelope Evolution
Current position: Data Scientist at Analytics@Purp1e, Mumbai, India

- Aug 2022–May 2023: Mr. Rion Glenn Nazareth, M.Sc., NISER
*Estimating Galactic Magnetic Fields from Observable Inputs:
 A Comprehensive Python Code for Semi-analytical Solutions*
 Current position: Ph.D. at Newcastle University, UK
- Aug 2022–May 2023: Mr. Soumik Bhattacharyya, M.Sc., NISER
Common Envelope Evolution: Towards Building a Simple Model
 Current position: M.Tech. at Indian Institute of Information Technology, Allahabad, India
- Aug 2018–Jun 2022: Ms. Yangyuxin Zou, Ph.D., University of Rochester (co-supervised with Adam Frank)
 Current position: Ph.D. at University of Rochester, USA
- Apr 2018–May 2020: Mr. Yisheng Tu, B.Sc. Research Student, University of Rochester
 Current position: Ph.D. at University of Virginia, USA

I have also acted as Thesis Coordinator for physics M.Sc. thesis students supervised by advisors in the Dept. of Earth and Planetary Sciences (7 students), Dept. of Computer Science (1) or external to NISER (1).

Contributions to Successful Proposals for Computational Resources/Funding

- Sep 2026: ACCESS Accelerate:
L. Chamandy, J. Nordhaus (PI), E. G. Blackman, B. Liu
Circumbinary Triple Common Envelope Evolution of Stars
 48,000 node hours requested on Stampede3 at Texas Advanced Computing Centre, USA.
 48,000 node hours awarded on Stampede3 from Sep 20, 2026 to Sep 19, 2027.
- Jul 2026: C-DAC PARAM Rudra:
L. Chamandy
Triple Common Envelope Phase of Stars
 603,000 node hours requested on PARAM Rudra at C-DAC, India.
 150,750 node hours awarded on PARAM Rudra for the period from Sep 1, 2026 to Nov 30, 2026.
- Oct 2023: Frontera Pathways:
L. Chamandy, E. G. Blackman, J. Nordhaus (PI), B. Liu
Simulating the Common Envelope Phase of Binary Stars up to Termination
 128,000 node hours requested on Frontera at Texas Advanced Computing Center, USA.
 100,000 node hours awarded on Frontera for the period from Dec 1, 2023 to Nov 30, 2024.
- Oct 2023: ACCESS Accelerate:
L. Chamandy, E. G. Blackman (PI), J. Nordhaus, B. Liu
Bi-polar Jet Launched from a Neutron Star Companion During Common Envelope Evolution
 2,485,700 core hours requested on Anvil at Purdue University, USA.
 2,485,700 core hours awarded from Nov 6, 2023 to Nov 5, 2024.
- Jan 2020: Extreme Science and Engineering Discovery Environment (XSEDE):
L. Chamandy, A. Frank, E. G. Blackman (PI), Y. Zou, B. Liu, J. Carroll-Nellenback
Common Envelope Evolution in Stars
 260,000 node hours requested on Stampede 2 at Texas Advanced Computing Center, USA.
 224,262 node hours awarded on Stampede 2 for the period from Apr 1, 2020 to Mar 31, 2021.
- Jan 2019: Extreme Science and Engineering Discovery Environment (XSEDE):
L. Chamandy, A. Frank (PI), E. G. Blackman, B. Liu, J. Carroll-Nellenback
Common Envelope Evolution in Stars
 250,000 node hours requested on Stampede 2 at Texas Advanced Computing Center, USA.
 190,341 node hours awarded on Stampede 2 for the period from Apr 1, 2019 to Mar 31, 2020.
- Jan 2018: National Science Foundation grant #1813298 (Galactic Astronomy Program, USA)
 “Interacting Binaries: Mass Transfer and Common Envelope Evolution”
 PIs: A. Frank, E. G. Blackman
 Amount awarded: \$291,367
 Start date: Sep 1, 2018

- Oct 2017: Extreme Science and Engineering Discovery Environment (XSEDE):
A. Frank (PI), B. Liu, **L. Chamandy**, J. Carroll-Nellenback
Mass Transfer in Evolved-Star Binary Systems: From Roche-Lobe Overflow to Bondi-Hoyle
230,000 node hours requested on Stampede 2 at Texas Advanced Computing Center, USA.
165,652 node hours awarded on Stampede 2 for the period from Jan 1 to Dec 31, 2018.
- Oct 2016: Extreme Science and Engineering Discovery Environment (XSEDE):
A. Frank (PI), J. Carroll-Nellenback, B. Liu, **L. Chamandy**
Mass Transfer in Evolved-Star Binary Systems: From Roche-Lobe Overflow to Bondi-Hoyle
4.2 million CPU hours requested on Stampede at TACC,
and 3.9 million CPU hours on Comet at San Diego Supercomputer Center, USA.
139,385 CPU hours awarded on Stampede and 980,222 CPU hours on Comet
for the period from Jan 1 to Dec 31, 2017.

Technical and Language Skills

- Programming: Proficiency with Fortran, Python, IDL, Git.
Software: Proficiency with ASTROBEAR, VISIT, MESA. Familiarity with MPI, PENCIL CODE.
Languages: English (native), French (fluent), Tamil (novice), Hindi (novice).

Code Development

- S. Ghosh & **L. Chamandy**, MAGNETIZER-RC, <https://ascl.net/2008.011>, DOI:10.5281/zenodo.21840438
Member, ASTROBEAR development team.
- L. F. S. Rodrigues & **L. Chamandy**, MAGNETIZER, <https://ascl.net/2008.011>, DOI:10.5281/zenodo.3817091
- R. G. Nazareth, G. Santhosh & **L. Chamandy**, GOBLIN, DOI:10.5281/zenodo.15118868
- R. G. Nazareth & **L. Chamandy**, SCALING-RELATIONS

Refereed Publications

1st Author or Primary Supervisor

34. D. Gurjar, **L. Chamandy**, E. G. Blackman, Y. Zou, B. Liu & J. Nordhaus 2026
Effect of Neutron Star Jets on Common Envelope Evolution
ApJ, in press (arXiv:2607.10267)
33. S. Ghosh, **L. Chamandy**, C. Jose, A. Shukurov, L. F. S. Rodrigues & F. S. Tabatabaei 2026
Radio Continuum Emission from Evolving Star-forming Galaxies I. Correlations Involving the Total Synchrotron Luminosity
ApJ 1009, 2 (arXiv:2606.01957)
32. S. Bhattacharyya, **L. Chamandy**, E. G. Blackman, A. Frank & B. Liu 2026
Understanding the Drag Torque in Common Envelope Evolution
PASA 43, e002 (arXiv:2506.19547)
30. R. G. Nazareth, G. Santhosh & **L. Chamandy**, 2025
Galactic Magnetic Fields. II. Applying the Model to Nearby Galaxies
ApJ 988, 197 (arXiv:2502.13029)
29. **L. Chamandy**, J. Nordhaus, E. G. Blackman & E. Wilson 2025
Second-generation Planet Formation After Tidal Disruption from Common Envelope Evolution
PASA 42, 27C (arXiv:2407.14190)
27. **L. Chamandy**, R. G. Nazareth & G. Santhosh 2024
Galactic Magnetic Fields. I. Theoretical Model and Scaling Relations
ApJ 966, 78 (arXiv:2311.15612)
26. **L. Chamandy**, J. Carroll-Nellenback, E. G. Blackman, A. Frank, Y. Tu, B. Liu, Y. Zou & J. Nordhaus 2024
How Negative Feedback and the Ambient Environment Limit the Influence of Recombination in Common Envelope Evolution
MNRAS 528, 234 (arXiv:2304.14840)

25. Y. Zou, **L. Chamandy**, J. Carroll-Nellenback, E. G. Blackman & A. Frank 2022
Jets from Main Sequence and White Dwarf Companions During Common Envelope Evolution
MNRAS 514, 304 (arXiv:2202.05715)
23. **L. Chamandy**, E. G. Blackman, J. Nordhaus & E. Wilson, 2021
Successive Common Envelope Events from Multiple Planetary Companions
MNRAS 502, L110 (arXiv:2011.11106)
22. **L. Chamandy** & A. Shukurov, 2020
Parameters of the Supernova-Driven Interstellar Turbulence
(Special Issue: “New Perspectives on Galactic Magnetism”, Eds: S. A. Mao & A. Fletcher)
Galaxies 8(3), 56 (arXiv:2007.14159)
20. **L. Chamandy**, E. G. Blackman, A. Frank, J. Carroll-Nellenback & Y. Tu, 2020
Common Envelope Evolution on the Asymptotic Giant Branch: Unbinding within a Decade?
MNRAS, 495, 4028 (arXiv:2004.06829)
17. **L. Chamandy**, E. G. Blackman, A. Frank, J. Carroll-Nellenback, Y. Zou & Y. Tu, 2019
How Drag Force Evolves in Global Common Envelope Simulations
MNRAS, 490, 3727 (arXiv:1908.06195)
15. **L. Chamandy**, Y. Tu, E. G. Blackman, J. Carroll-Nellenback, A. Frank, B. Liu & J. Nordhaus, 2019
Energy Budget and Core-Envelope Motion in Common Envelope Evolution
MNRAS, 486, 1070 (arXiv:1812.11196)
Erratum: MNRAS, 501, 2209 (2021)
13. **L. Chamandy** & N. K. Singh, 2018
Non-linear Galactic Dynamos and the Magnetic Radler Effect
MNRAS, 481, 1300 (arXiv:1805.05548)
12. **L. Chamandy**, A. Frank, E. G. Blackman, J. Carroll-Nellenback, B. Liu, Y. Tu, J. Nordhaus, Z. Chen & B. Peng, 2018
Accretion in Common Envelope Evolution
MNRAS, 480, 1898 (arXiv:1805.03607)
10. **L. Chamandy** & N. K. Singh, 2017
A New Constraint on Mean-field Galactic Dynamo Theory
MNRAS, 468, 3657 (arXiv:1612.05289)
9. **L. Chamandy**, A. Shukurov & A. R. Taylor, 2016
Statistical Tests of Galactic Dynamo Theory
ApJ, 833, 43 (arXiv:1609.05688)
8. **L. Chamandy**, 2016
An Analytical Dynamo Solution for Large-scale Magnetic Fields of Galaxies
MNRAS, 462, 4402 (arXiv:1608.01119)
7. **L. Chamandy** & A. R. Taylor, 2015
Non-linear Galactic Dynamos and the Magnetic Pitch Angle
ApJ, 808, 28 (arXiv:1506.03245)
6. **L. Chamandy**, A. Shukurov & K. Subramanian, 2015
Magnetic Spiral Arms and Galactic Outflows
MNRAS, 446, L6 (arXiv:1408.3937)
5. **L. Chamandy**, A. Shukurov, K. Subramanian & K. Stoker, 2014
Non-linear Galactic Dynamos: A Toolbox
MNRAS, 443, 1867 (arXiv:1403.2562)
4. **L. Chamandy**, K. Subramanian & A. Quillen, 2014
Magnetic Arms Generated by Multiple Interfering Galactic Spiral Patterns
MNRAS, 437, 562 (arXiv:1308.0432)
3. **L. Chamandy**, K. Subramanian & A. Shukurov, 2013
Galactic Spiral Pattern and Dynamo Action II: Asymptotic Solutions
MNRAS, 433, 3274 (arXiv:1301.4761)

2. **L. Chamandy**, K. Subramanian & A. Shukurov, 2013
Galactic Spiral Patterns and Dynamo Action I: A New Twist on Magnetic Arms
MNRAS, 428, 3569 (arXiv:1207.6239)

Other Refereed Publications

31. I. Paul, R. V. Kashyap, T. Ghosh, R. Beck, **L. Chamandy**, S. Sinha & A. Shukurov 2026
The Fine Structure of the Mean Magnetic Field in M31
MNRAS 545, 2041 (arXiv:2502.13029)
28. C. Jose, **L. Chamandy**, A. Shukurov, K. Subramanian, L. F. S. Rodrigues, C. Baugh 2024
Understanding the Radio Luminosity Function of Star-forming Galaxies and its Cosmological Evolution
MNRAS 532, 1504 (arXiv:2402.15099)
24. G. Guidarelli, J. Nordhaus, J. Carroll-Nellenback, **L. Chamandy**, A. Frank, E. G. Blackman, 2022
The Formation of Discs in the Interior of AGB Stars from the Tidal Disruption of Planets and Brown Dwarfs
MNRAS, 511, 5994 (arXiv:2105.00077)
21. Y. Zou, A. Frank, Z. Chen, T. Reichardt, O. De Marco, E. G. Blackman, J. Nordhaus, B. Balick, J. Carroll-Nellenback, **L. Chamandy** & B. Liu, 2020
Bipolar Planetary Nebulae from Outflow Collimation by Common Envelope Evolution
MNRAS, 497, 2855 (arXiv:1912.01647)
19. T. Reichardt, O. De Marco, R. Iaconi, **L. Chamandy** & D. Price, 2020
The impact of recombination energy on simulations of the common envelope binary interaction
MNRAS, 494, 5333 (arXiv:1911.02759)
18. R. Beck, **L. Chamandy**, E. Elson & E. G. Blackman, 2019
Synthesizing Observations and Theory to Understand Galactic Magnetic Fields: Progress and Challenges
(Special Issue: “New Perspectives on Galactic Magnetism”, Eds: S. A. Mao & A. Fletcher)
Galaxies 8(1), 4 (arXiv:1912.08962)
16. G. Guidarelli, J. Nordhaus, **L. Chamandy**, Z. Chen, E. G. Blackman, A. Frank, J. Carroll-Nellenback & B. Liu, 2019
Hydrodynamic Simulations of Disrupted Planetary Accretion Disks Inside the Core of an AGB Star
MNRAS, 490, 1179 (arXiv:1908.00157)
14. L. F. S. Rodrigues, **L. Chamandy**, A. Shukurov, C. M. Baugh & A. R. Taylor, 2019
Evolution of Galactic Magnetic Fields
MNRAS, 483, 2424 (arXiv:1809.10521)
11. H. Zhou, E. G. Blackman & **L. Chamandy**, 2018
Derivation and Precision of Mean Field Electrodynamics with Mesoscale Fluctuations
Journal of Plasma Physics, 84, 3 (Special issue: *50 Years of Mean Field Electrodynamics*)
(arXiv:1710.04064)
1. M. Ishak, **L. Chamandy**, N. Neary & K. Lake, 2001
Exact Solutions with W-modes
PhRvD 64, 024005 (gr-qc/0007073)

Conference Proceedings/Abstracts

- S. Ghosh, **L. Chamandy**, C. Jose & L. F. S. Rodrigues, 2025
Evolution of Galactic Magnetic Fields and Scaling Relations in Radio Continuum
43rd Meeting of the Astronomical Society of India, 2025
- D. Gurjar, **L. Chamandy**, A. Zou, E. G. Blackman, B. Liu & J. Nordhaus 2024
Impact of Neutron Star Jet on Common Envelope Evolution
European Astronomical Society Annual Meeting 2024
- L. Chamandy**, Y. Tu, J. Carroll-Nellenback, E. G. Blackman, A. Frank, B. Liu, Y. Zou, J. Nordhaus, 2023
Recombination and Ionization During Common Envelope Evolution and the Role of the Environment
European Astronomical Society Annual Meeting 2023

- L. Chamandy**, A. Frank, E. G. Blackman, J. Carroll-Nellenback, B. Liu, Y. Tu, J. Nordhaus, Z. Chen & B. Peng, 2019
Accretion in Common Envelope Evolution
 IAU Symposium 343, 235 (arXiv:1810.04757)
- L. Chamandy**, A. Shukurov & A. R. Taylor, 2018
New Insights on Galactic Dynamos
 IAU Focus Meeting 8 (arXiv:1810.07246)
- L. Chamandy**, K. Subramanian & A. Shukurov, 2013
Galactic Spiral Patterns and Dynamo Action
 IAU Symposium 294, 249
- M. Ishak, **L. Chamandy** & K. Lake, 2001
Exact Solutions with W-modes: Trapping of Gravitational Waves Inside Neutron Stars
 American Institute of Physics Conference Series 586, 550

Conference Presentations

Invited Review Talks

- May 2019 *Review of Numerical Simulations of Common Envelope Evolution*
 Common Envelope Evolution 2019
 Center for Computational Astrophysics, Flatiron Institute, New York, United States

Other Conference Talks

- Jan 2026 *Understanding the Radio Luminosity Function of Star-forming Galaxies and its Cosmological Evolution*
 SKA-India Science Meeting,
 IISER Mohali, India
- Jun 2025 *Understanding the Drag Torque in Common Envelope Evolution*
 European Astronomical Society Annual Meeting,
 Cork, Ireland
- May 2024 *Galactic Magnetic Fields: A Theoretical Perspective*
 Cosmic Magnetism in the Pre-SKA Era,
 Kagoshima, Japan
- Jul 2023 *How Negative Feedback Limits the Influence of Recombination in Common Envelope Evolution*
 European Astronomical Society Annual Meeting,
 Krakow, Poland
- Jun 2022 *Global Simulations of Common Envelope Evolution with Companion Jets*
 Physics and Astrophysics of Common Envelopes
 Los Alamos, New Mexico
- Oct 2021 *Evolution of Galactic Magnetic Fields*
 Royal Astronomical Society Specialist Discussion Meeting on Galactic Magnetic Fields (recorded)
- Oct 2021 *Global Simulations of Common Envelope Evolution with Companion Jets*
 Asymmetrical Post-main-sequence Nebulae 8e2021 (APN8e) (talk given remotely)
- Oct 2021 *Toward Theoretical Scaling Relations for Magnetic Fields in the Milky Way and Other Spiral Galaxies*
 IMAGINE Workshop: Modeling the Galactic Magnetic Field (talk given remotely)
- Sep 2021 *Common Envelope Evolution and Low-mass Companions*
 Common Envelope Physics and Outcomes (CEPO2021) (talk given remotely)
- Jul 2020 *An Analytical Dynamo Model for the Large-scale Magnetic Field in a Galactic Disc*
 The Interstellar Magnetic Field Inference Engine Consortium Workshop
 Leiden, The Netherlands (talk given remotely)
- Jul 2020 *Simulating Common Envelope Evolution Involving an Asymptotic Giant Branch Primary*
 EAS Ann. Meeting Sympos. 8: Common envelope systems: progenitors, mergers and survivors
 Leiden, The Netherlands (talk given remotely)

- Jul 2019 *Evolution of Galactic Magnetic Fields*
Modeling MeerKATs: Comparing galaxy formation simulations to MeerKAT surveys
Kruger Park, South Africa
- Jul 2019 *Energy Budget and Drag Force in 3D Global AMR Simulations of Common Envelope Evolution*
The Beginning and Ends of Double White Dwarfs
Copenhagen, Denmark
- Jun 2019 *Synthesizing Theory and Observation to Understand Galactic Magnetic Fields*
New Perspectives on Galactic Magnetism
Newcastle, United Kingdom
- Aug 2018 *New Insights on Galactic Dynamos*
IAU Focus Meeting 8: New Insights in Extragalactic Magnetic Fields
Vienna, Austria
- Aug 2018 *Accretion in Common Envelope Evolution*
IAU Symposium 343: Why galaxies care about AGB stars
Vienna, Austria
- Aug 2017 *Effects of Small-scale Magnetic Fields on Large-scale Galactic Dynamos*
Plasma Universe and its Structure Formation
Inter-University Centre for Astronomy and Astrophysics, Pune, India
- Jul 2017 *Simulations of Common Envelope Evolution with AstroBEAR: First Results*
The Physics of Evolved Stars II: The Role of Binarity
Nice, France
- Apr 2016 *Simulating the Cosmic Evolution of Magnetic Fields*
Institute for Data-Intensive Astronomy Data Science Workshop
University of the Western Cape, Cape Town, South Africa
- Jun 2015 *Non-linear Galactic Dynamos and Magnetic Pitch Angles*
Origin, Evolution, and Signatures of Cosmological Magnetic Fields
Nordita, Stockholm
- Dec 2014 *Dynamo Action in Spiral Galaxies*
Square Kilometre Array Bursary Conference
Stellenbosch, South Africa
- Oct 2014 *Dynamo Action in Spiral Galaxies*
Cosmic Magnetic Fields: Current Knowledge and Future Ideas
Krakow, Poland
- Sep 2013 *Building Magnetic Arms with Large-scale Dynamos*
Dynamics of Earth and Planetary Cores
Centre for Earth Sciences, Indian Institute of Science, Bangalore
- Jun 2012 *Galactic Spiral Patterns and Dynamo Action: A New Twist on Magnetic Arms*
East-Asia School and Workshop on Laboratory, Space and Astrophysical Plasmas
Jeju, South Korea

Posters

- Apr 2019 *Global 3D Simulations of Common Envelope Evolution using AMR*
Spring Symposium: The Deaths and Afterlives of Stars
Space Telescope Science Institute, Baltimore, United States
- Aug 2012 *Galactic Spiral Patterns and Dynamo Action: A New Twist on Magnetic Arms*
IAU Symposium 294, Solar and Astrophysical Dynamos
Beijing, China

Seminars and Colloquia

- Sep 2026 *Global Hydrodynamical Simulations of Common Envelope Evolution in 3D and 1D*
Physics Seminar
Institute of Mathematical Sciences, India

- Sep 2026 *Predicting Radio Continuum Population Statistics of Evolving Star-Forming Galaxies using MAGNETIZER*
Astrophysics Seminar
Raman Research Institute, India
- May 2025 *Second-generation Planet Formation After Tidal Disruption from Common Envelope Evolution*
Astro seminar
Université de Montréal, Canada
- Jun 2023 *What we have Learned from 3D Hydrodynamical Simulations of Common Envelope Evolution*
Center for Computational Relativity and Gravitation Seminar
Rochester Institute of Technology, USA
- Jun 2023 *What we have Learned from 3D Hydrodynamical Simulations of Common Envelope Evolution*
Astro seminar
Université de Montréal, Canada
- May 2023 *Common Envelope Evolution: What we are Learning from 3D Global Simulations*
Online astrophysics seminar, Hamburg Observatory, Germany (talk given remotely)
- Mar 2023 *Testable Galactic Dynamo Models*
Astrophysics Seminar
Indian Institute of Astrophysics, Bengaluru, India
- Jun 2021 *Successive Common Envelope Events from Multiple Planets*
Online Meetings on Evolved Stars and Systems (O-MESS) (talk given remotely)
- Apr 2021 *Common Envelope Evolution in Stars and Magnetic Field Evolution in Galaxies*
Physics Seminar
Valdosta State University, Valdosta, GA, USA
- Aug 2020 *Magnetic Field Evolution in Galaxies and Common Envelope Evolution in Stars*
Physics Seminar (talk given remotely)
Institute of Mathematical Sciences, Chennai, India
- Aug 2020 *Magnetic Field Evolution in Galaxies and Common Envelope Evolution in Stars*
Seminar (talk given remotely)
Indian Institute of Astrophysics, Bengaluru, India
- Jun 2020 *Magnetic Field Evolution in Galaxies and Common Envelope Evolution in Stars*
Research Seminar (talk given remotely)
National Institute of Science Education, Bhubaneswar, India
- May 2020 *Magnetic Field Evolution in Galaxies and Common Envelope Evolution in Stars*
Instituto de Astronomía, Universidad Nacional Autónoma de México
Mexico city, Mexico (talk given remotely)
- Oct 2019 *Simulating Common Envelope Evolution*
Center for Integrated Research Computing Symposium
University of Rochester, Rochester, NY, USA
- Jun 2019 *Global 3D Simulations of Common Envelope Evolution: Progress, Prospects & Perspectives*
Physics Seminar
Newcastle University, Newcastle upon Tyne, United Kingdom
- Aug 2018 *Galactic Dynamos: Toward Statistical Tests of Dynamo Models*
MPS Seminar
Max Planck Institute for Solar System Research, Göttingen, Germany
- Jul 2018 *Accretion in Common Envelope Evolution*
Astrophysics Seminar
Inter-University Centre for Astronomy and Astrophysics, Pune, India
- Nov 2017 *Common Envelope Simulations and (Separately) Exploring a New Dynamo Effect*
Astro Seminar
Université de Montréal, Montreal, Canada
- Jul 2017 *Toward Statistical Tests of Galactic Dynamo Theory*
MPIfR Colloquium
Max Planck Institute for Radio Astronomy, Bonn, Germany

- Jul 2017 *Exploring a New Effect in Large-scale Galactic Dynamos*
MPS Seminar,
Max Planck Institute for Solar System Research, Göttingen, Germany
- Mar 2017 *Dynamos in Spiral Galaxies: A New Constraint from Strong Magnetic Fluctuations*
Astro Lunch Seminar
Newcastle University, United Kingdom
- Jan 2017 *Explaining the Global Properties of Spiral Galaxy Magnetic Fields*
Physics & Astronomy Seminar
University of Rochester, Rochester, NY, United States
- Jun 2016 *Large-scale Dynamo Action in Galaxies: Statistical Tests of Galactic Dynamo Theory*
Applied Maths & Physics Seminar
Newcastle University, United Kingdom
- Jan 2016 *Pitch Angles of Magnetic Fields of Galaxies: Toward Statistical Tests of Dynamo Theory*
Astrophysics Seminar
Inter-University Centre for Astronomy and Astrophysics, Pune, India
- Oct 2015 *Galactic Dynamos and the Magnetic Pitch Angle*
Astro Lunch Seminar
Newcastle University, United Kingdom
- Jul 2014 *Dynamo Action in Spiral Galaxies*
Physics Seminar
University of the Western Cape, Cape Town, South Africa
- Jul 2014 *Dynamo Action in Spiral Galaxies*
Astronomy Seminar
University of Cape Town, Cape Town, South Africa
- May 2014 *Dynamo Action in Spiral Galaxies*
Physics and Astronomy Special Seminar
University of Rochester, Rochester, NY, United States
- Apr 2014 *Dynamo Action in Spiral Galaxies*
Astronomy and Astrophysics Seminar
Université de Montréal, Montreal, Canada
- Apr 2014 *Dynamo Action in Spiral Galaxies*
Physics, Engineering Physics and Astronomy Colloquium
Queen's University, Kingston, Canada

Honours/Awards

- 2020 Selected for Editor's Choice, *Galaxies*,
Parameters of the Supernova-Driven Interstellar Turbulence
- 2018 Feature article, *Journal of Plasma Physics*
Derivation and Precision of Mean Field Electrodynamics with Mesoscale Fluctuations
- 2018 International Astronomical Union Travel Grant
- 2012 International Astronomical Union Travel Grant
- 2003 Natural Sciences and Engineering Research Council of Canada Research Scholarship (declined)
- 2001–2002 Ontario Graduate Scholarship
- 1999–2001 Senator Frank Carrel Upper Year Scholarship, Queen's University

Organization and Coordination of Academic Events

- 2024–present Spearheaded and coordinated a research training program for M.Sc. students, NISER
Consists of four interactive sessions for each graduating class
(taking place during semesters 8 and 9 of their 10 semester program)
Topics are (1) How to coordinate an M.Sc. Project, (2) Applying for PhD positions,
(3) Strategies for writing a report/thesis/paper, (4) How to give a talk.

2026	Head Organizer for Galactic and Extragalactic Magnetism Seminars (GEMS) online seminar series To begin in October 2026: https://sites.google.com/niser.ac.in/gems/home .
2023	Co-organizer (main organizer Tuhin Ghosh) <i>Advanced 21 cm Cosmology School and Workshop</i> Conference held at NISER on Dec. 11-21, 2023
2023	Organizer (jointly with Narayan Rana) <i>School of Physical Sciences Day</i> Conference held at NISER on Mar. 26-27, 2023
2021	Co-organizer (jointly with Amit Seta) <i>Royal Astronomical Society Specialist Discussion Meeting on Galactic Magnetic Fields</i> Conference held online on Oct. 8, 2021
2017	Co-organizer (jointly with Pallavi Bhat) <i>Plasma Universe and its Structure Formation</i> Conference held at the Inter-university Centre for Astronomy & Astrophysics, Pune, India
2014–2016	Organizer of Graduate Student Seminars Astronomy Department, University of Cape Town

Committee Membership

2022–present	UGCS committee member, School of Physical Sciences, NISER (deals with all matters related to Integrated B.Sc.-M.Sc. physics program). I have taken a very active role in this committee and have spearheaded several new initiatives, such as a new evaluation scheme for M.Sc. projects.
2024–present	Coordinator of Astrophysics and Cosmology faculty hiring committee

Professional Affiliations

2026–present	Member of the International Astronomical Union
2023–present	Member of the SKA-India Science Working Group
2022–present	Associate, Inter-University Centre for Astronomy and Astrophysics
2020–present	Member of the SKA Magnetism Science Working Group
2015–2020	Research Associate, School of Mathematics, Statistics & Physics, Newcastle University, U.K.

Review Work

2017–present	Referee, <i>Monthly Notices of the Royal Astronomical Society</i>
2015–present	Referee, <i>The Astrophysical Journal</i>
2020	Referee, <i>Nature Astronomy</i>
2019–2020	Referee, <i>Galaxies</i>
2019	Referee, <i>European Research Council Advanced Grant Proposal</i>
2016	Referee, Giant Metrewave Radio Telescope observing proposal
Oct 2014	Judge for student talks and poster presentations Square Kilometre Array Bursary Conference, Stellenbosch, South Africa

Scientific Outreach

2022–Present	Faculty supervisor (jointly with three others) for the NISER Astronomy Club
2023–2025	Annual astrophysics presentation/interaction sessions with school students KIIT International School, Bhubaneswar
Aug 2024	Presented “International Space Day” talk to school students, NISER

Nov 2019	Presenter of talk about exoplanets to 8th grade science students Twelve Corners Middle School, Rochester, New York
2015	Mentor in Job Shadowing Programme Astronomy Department, University of Cape Town
2015	Presenter of talks about astronomy to primary school students Sea Point Primary School, Cape Town, South Africa
2013–2014	Contributor to <i>History of Astronomy Poster Project</i> , Inter-University Centre for Astronomy and Astrophysics, Pune, India
Feb 2013	Presenter of public talk <i>The Search for Planets Outside of our Solar System</i> , Inter-University Centre for Astronomy and Astrophysics, Pune, India
2010–2013	Presenter in Astronomy & Astrophysics Poster Exhibition for Science Day, Inter-University Centre for Astronomy and Astrophysics, Pune, India
Feb 2011	Conductor of Science Day Quiz for the Public, Inter-University Centre for Astronomy and Astrophysics, Pune, India
2009–2011	Presenter of annual astrophysics talks as part of “Faculty Lecture Series”, Mahindra United World College of India, Pune, India
2010	Contributor to Science Day Quiz for School Students, Inter-University Centre for Astronomy and Astrophysics, Pune, India

References

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